Econometrics I
Problem Set 2: OLS applications and variance

Matías Cabello
University Halle-Wittenberg
Chair of Economic Growth and Development

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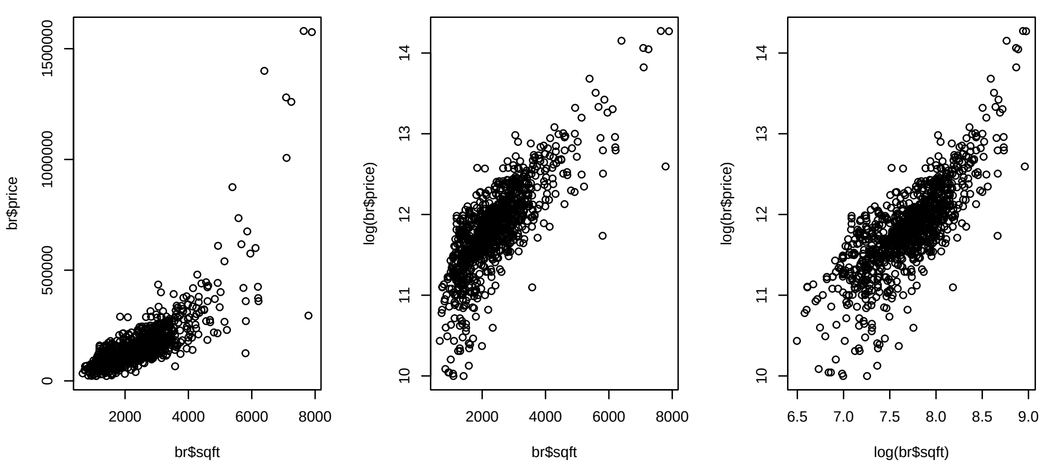
1. **Application: real estate data.** Download `br.RData` from StudIP. This dataset (originally from Hill et al. 2018, *Principles of Econometrics*) contains data on 1080 houses sold in Baton Rouge, Louisiana, during mid-2005.

   (a) Plot `price` vs. `sqft`. What kind of model or transformation would the scatter suggest?

   Solution:

   The level-level plot is clearly non-linear. This calls for either a polynomial model or a transformation. The log-linear and log-log plots suggest that the latter transformation is more adequate.

   (b) Estimate the relationship of `price` vs. `sqft` with a level-level, a quadratic polynomial, a log-level, and a log-log model. You should get the results shown in table 1. What model offers the best fit? Does your graphical analysis lead to the same conclusion?

Table 1: Result of question 1b

	Dependent variable:			
	price		log(price)	
	(1)	(2)	(3)	(4)
sqft	92.747^{***}	-41.153^{***}	0.0004^{***}	
	(2.411)	(6.623)	(0.00001)	
I(sqftˆ2)		0.021^{***}		
		(0.001)		
log(sqft)				1.006^{***}
				(0.025)
Constant	$-60{,}861.460^{***}$	$113{,}226.500^{***}$	10.839^{***}	4.080^{***}
	(6,110.187)	(9,672.195)	(0.025)	(0.189)
Observations	1,080	1,080	1,080	1,080
R^2	0.579	0.703	0.625	0.607
Note:				*p<0.1; **p<0.05; ***p<0.01

Solution: According to the R^2, the best model is the quadratic one. However, visual inspection reveals that all but the log-log seem sensitive to outliers. Therefore, the recommended one would be the log-log specification.

(c) You are considering buying a 2,000 square-feet property for \$ 120,000. Is that a good price? Answer according to each estimate of table 1.

Solution:

	(1)	(2)	(3)	(4)
x	2000	2000	2000	2000
$\hat{y}$	\$124,633	\$114,921	\$113,437	\$123,811
Gain/loss	\$4,633	-\$5,080	-\$6,563	\$3,811

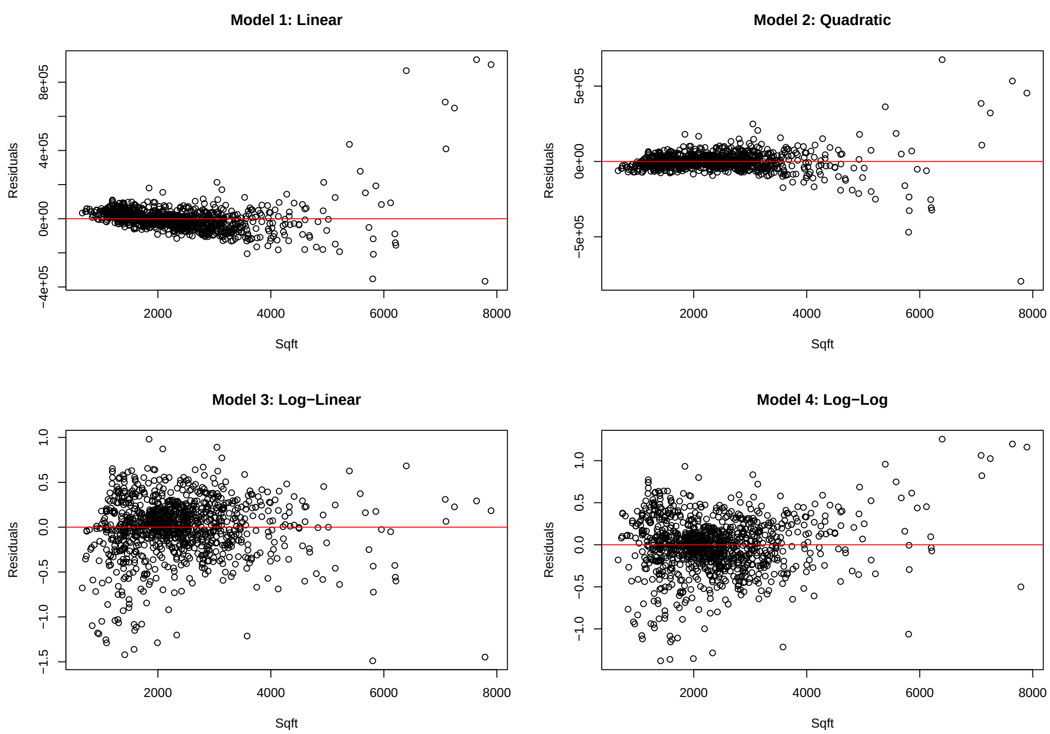
Models 2 and 3 suggest that the price is above average. Models 1 and 4 suggest the opposite.

(d) Interpret the effect of size on price according to each estimate of of table 1.

Solution: (1) An extra square feet increase the price by \$ 92.75
(2) An extra square feet increase the price by $-41.153 + 0.021 \cdot 2 \cdot sqft$ \$ USD.
(3) An extra square feet increase the price by 0.04%.
(4) A 1% increase in size increases the price by 1.006%

(e) Do the regressions exhibit homoskedastic errors? Generate the residuals via, e.g., `br$resid1 <- residuals(model1)` and plot them against `sqft`. What is no longer valid in case of heteroskedasticity?

Solution:



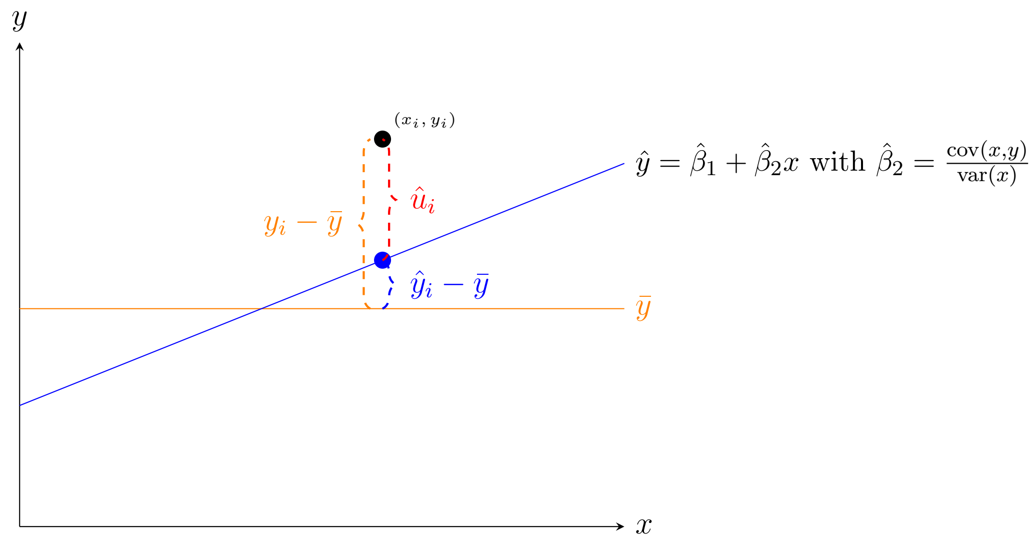
Models 1 and 2 are clearly heteroskedastic, 4 also to some extent. This means that the reported st. errors of the coefficients $\hat{s}(\hat{\beta}_j)$ are not properly computed and cannot be trusted. Only 3 appears to obey homoskedasticity.

2. **R-squared**

(a) Explain graphically why the variance of y can be decomposed as the variance of $\hat{y}$

and that of $\hat{u}$.

Solution:



(b) Show that

$$\underbrace{\sum_{i=1}^{n}(y_i - \bar{y})^2}_{TSS} = \underbrace{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}_{ESS} + \underbrace{\sum_{i=1}^{n}\hat{u}_i^2}_{SSR}$$

so that $0 \leq R^2 \leq 1$.

Solution:

Recall the two first order conditions $\frac{\partial SSR}{\partial \hat{\beta}_1} = 0$ and $\frac{\partial SSR}{\partial \hat{\beta}_1} = 0$ from problem set 1. They can be rewritten as

$$\sum_{i=1}^{n}\hat{u}_i = 0 \quad (1^*)$$

$$\sum_{i=1}^{n}x_i\hat{u}_i = 0 \quad (2^*)$$

We will use them below.

Let us expand TSS as follows:

$$\begin{aligned}
\text{TSS} &= \sum_{i=1}^{n}(y_i - \bar{y})^2 \\
&= \sum_{i=1}^{n}(\hat{y}_i - \bar{y} + \hat{u}_i)^2 \quad (\text{since } y_i = \hat{y}_i + \hat{u}_i) \\
&= \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 + \sum_{i=1}^{n}\hat{u}_i^2 + 2\sum_{i=1}^{n}(\hat{y}_i - \bar{y})\hat{u}_i
\end{aligned}$$

Now examine the cross term:

$$\begin{aligned}
\sum_{i=1}^{n}(\hat{y}_i - \bar{y})\hat{u}_i &= \sum_{i=1}^{n}\hat{y}_i\hat{u}_i - \bar{y}\sum_{i=1}^{n}\hat{u}_i \\
&= \sum_{i=1}^{n}(\hat{\beta}_1 + \hat{\beta}_2 x_i)\hat{u}_i - 0 \quad (\text{by } (1^*)) \\
&= \hat{\beta}_1\sum_{i=1}^{n}\hat{u}_i + \hat{\beta}_2\sum_{i=1}^{n}x_i\hat{u}_i \\
&= 0 + 0 = 0 \quad (\text{by } (1^*) \text{ and } (2^*))
\end{aligned}$$

Therefore:

$$\begin{aligned}
\text{TSS} &= \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 + \sum_{i=1}^{n}\hat{u}_i^2 \\
&= \text{ESS} + \text{RSS}
\end{aligned}$$

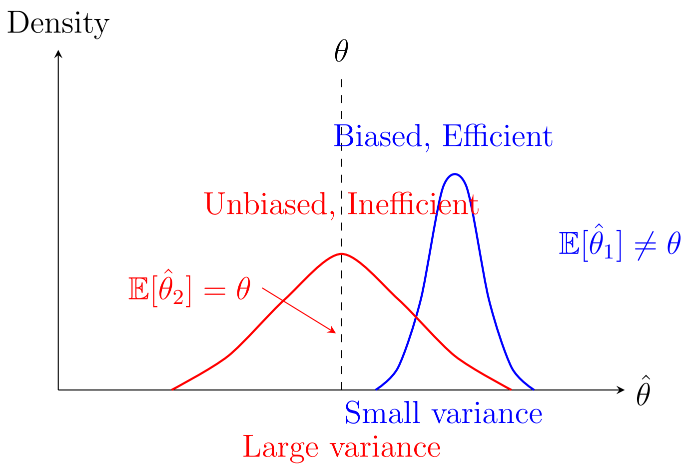
And R^2 is defined as:

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$$

3. **Unbiasedness**. We say that an estimator $\hat{\theta}$ of the population parameter θ is unbiased if, on average in repeating sampling, it coincides with its population counterpart—i.e., when $\mathbb{E}(\hat{\theta}) = \theta$.

   (a) Distinguish graphically between unbiasedness and efficiency.

Solution:



(b) Starting from the formula

$$\hat{\beta}_2 = \frac{\text{cov}(x, y)}{\text{var}(x)}$$

of the simple (univeariate) regression model, show that the condition $\mathbb{E}(u_i|x) = 0$—or, alternatively, $\mathbb{E}[\text{cov}(u, x)] = 0$—is required in addition to the assumption

$$y = \beta_1 + \beta_2 x + u$$

for $\mathbb{E}(\hat{\beta}_2|x) = \beta_2$ to hold.

Solution: By the rules described in the lecture's appendix:

$$\begin{aligned}
\hat{\beta}_2 &= \frac{\text{cov}(x, x\beta_2)}{\text{var}(x)} + \frac{\text{cov}(x, u)}{\text{var}(x)} \\
&= \beta_2 \frac{\text{cov}(x, x)}{\text{var}(x)} + \frac{\text{cov}(x, u)}{\text{var}(x)} \\
&= \beta_2 \frac{\text{var}(x)}{\text{var}(x)} + \frac{\text{cov}(x, u)}{\text{var}(x)} \\
&= \beta_2 + \frac{\text{cov}(x, u)}{\text{var}(x)} = \beta_2 + \frac{\sum_{i=1}^n (x_i - \bar{x}) u_i}{\sum_{i=1}^n (x_i - \bar{x})^2}
\end{aligned}$$

Hence, we get $\mathbb{E}(\hat{\beta}_2) = \beta_2$ by assuming $\mathbb{E}[\text{cov}(u, x)] = 0$ or by assuming $\mathbb{E}(u_i|x) = 0$, so that $\mathbb{E}\left(\frac{\sum_{i=1}^n (x_i - \bar{x}) u_i}{\sum_{i=1}^n (x_i - \bar{x})^2} | x\right) = \frac{\sum_{i=1}^n (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \mathbb{E}\left(u_i | x\right) = 0$.

(c) Starting from the formula

$$\hat{\beta} = (X'X)^{-1} X'y$$

of the multivariate regression model, show that the condition $\mathbb{E}(u|X) = 0$—or, alternatively, $\mathbb{E}(X'u) = 0$—is required in addition to the assumption

$$y = X\beta + u$$

for $\mathbb{E}(\hat{\beta}|X) = \beta$ to hold.

Solution:

$$\begin{aligned}
\hat{\beta} &= (X'X)^{-1}X'y \\
&= (X'X)^{-1}X'(X\beta + u) && \text{given } y = X\beta + u \\
&= \beta + (X'X)^{-1}X'u \\
\mathbb{E}(\hat{\beta}|X) &= \beta + \mathbb{E}[(X'X)^{-1}X'u|X] \\
&= \beta + (X'X)^{-1}X'\mathbb{E}[u|X] \\
&= \beta && \text{given } \mathbb{E}[u|X] = 0 \text{ or } \mathbb{E}[X'u] = 0
\end{aligned}$$

4. **Estimator variance**. We define an estimator variance as $\mathbb{E}[(\hat{\theta} - \mathbb{E}(\hat{\theta}))^2]$ in repeating sample.

   (a) Show that the variance of the slope $\hat{\beta}_2$ in a simple regression that we obtained in class,

$$\text{var}(\hat{\beta}_2|x) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, \tag{1}$$

   can also be obtained from solving $\mathbb{E}(\hat{\beta}_2 - \mathbb{E}(\hat{\beta}_2))^2$.

Solution:

$$\begin{aligned}
\text{var}(\hat{\beta}_2|x) &= \mathbb{E}[(\hat{\beta}_2 - \mathbb{E}[\hat{\beta}_2|x])^2|x] \\
&= \mathbb{E}[(\hat{\beta}_2 - \beta_2)^2|x] && \text{[from (b) we know:}\mathbb{E}[\hat{\beta}_2|x] = \beta_2] \\
&= \mathbb{E}\left[\left(\frac{\sum_{i=1}^n (x_i - \bar{x})u_i}{\sum_{i=1}^n (x_i - \bar{x})^2}\right)^2 \middle| x\right] && \text{[also from question (b)]} \\
&= \frac{1}{\left(\sum_{i=1}^n (x_i - \bar{x})^2\right)^2}\mathbb{E}\left[\left(\sum_{i=1}^n (x_i - \bar{x})u_i\right)^2 \middle| x\right]
\end{aligned}$$

Now expand the square:

$$\begin{aligned}
\mathbb{E}\left[\left(\sum_{i=1}^n (x_i - \bar{x})u_i\right)^2 \middle| x\right] &= \mathbb{E}\left[\sum_{i=1}^n \sum_{j=1}^n (x_i - \bar{x})(x_j - \bar{x})u_i u_j \middle| x\right] \\
&= \sum_{i=1}^n \sum_{j=1}^n (x_i - \bar{x})(x_j - \bar{x})\mathbb{E}[u_i u_j|x]
\end{aligned}$$

Note that, since $\mathbb{E}[u_i|x] = 0$ for all i, the following holds:

$$\text{cov}(u_i, u_j|x) = \mathbb{E}[(u_i - \mathbb{E}[u_i|x])(u_i - \mathbb{E}[u_i|x])|x] = \mathbb{E}[u_i u_i|x]$$

Therefore:

- $\mathbb{E}[u_i u_j|x] = \text{cov}(u_i, u_j|x) = 0$ for $i \neq j$ (no autocorrelation)
- $\mathbb{E}[u_i^2|x] = \text{var}(u_i|x) = \sigma^2$ (homoskedasticity)

Using these expressions, the expanded squared term has expectation

$$\begin{aligned}
\mathbb{E}\left[\left(\sum_{i=1}^{n}(x_i - \bar{x})u_i\right)^2 \middle| x\right] &= \sum_{i=1}^{n}(x_i - \bar{x})^2 \mathbb{E}[u_i^2|x] \\
&= \sigma^2 \sum_{i=1}^{n}(x_i - \bar{x})^2
\end{aligned}$$

Therefore, going back to our variance:

$$\begin{aligned}
\text{var}(\hat{\beta}_2|x) &= \frac{1}{\left(\sum_{i=1}^{n}(x_i - \bar{x})^2\right)^2}\mathbb{E}\left[\left(\sum_{i=1}^{n}(x_i - \bar{x})u_i\right)^2 \middle| x\right] \\
&= \frac{1}{\left(\sum_{i=1}^{n}(x_i - \bar{x})^2\right)^2}\sigma^2 \sum_{i=1}^{n}(x_i - \bar{x})^2 \\
&= \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}
\end{aligned}$$

(b) Show that, for a random vector $a_{(n\times 1)}$, it holds that

$$\underset{n\times n}{V[a]} \equiv \mathbb{E}\left[(a - \mathbb{E}[a])(a - \mathbb{E}[a])'\right] = \begin{bmatrix} \text{var}[a_1] & \text{cov}[a_1, a_2] & \cdots & \text{cov}[a_1, a_n] \\ \text{cov}[a_2, a_1] & \text{var}[a_2] & \cdots & \text{cov}[a_2, a_n] \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}[a_n, a_1] & \text{cov}[a_n, a_2] & \cdots & \text{var}[a_n] \end{bmatrix}$$

Note: we call $V[a]$ the **variance-covariance matrix** of a.

Solution:

$$
\begin{aligned}
\underset{n\times n}{V[a]} &= \mathbb{E}\left[(a-\mathbb{E}[a])(a-\mathbb{E}[a])'\right] \\
&= \mathbb{E}\left[\begin{bmatrix}(a_1-\mathbb{E}[a_1])\\ \vdots \\ (a_n-\mathbb{E}[a_n])\end{bmatrix}\begin{bmatrix}(a_1-\mathbb{E}[a_1]) & \dots & (a_n-\mathbb{E}[a_n])\end{bmatrix}\right] \\
&= \begin{bmatrix}\mathbb{E}\left[(a_1-\mathbb{E}[a_1])(a_1-\mathbb{E}[a_1])\right] & \cdots & \mathbb{E}\left[(a_1-\mathbb{E}[a_1])(a_n-\mathbb{E}[a_n])\right] \\ \mathbb{E}\left[(a_2-\mathbb{E}[a_2])(a_1-\mathbb{E}[a_1])\right] & \cdots & \mathbb{E}\left[(a_2-\mathbb{E}[a_2])(a_n-\mathbb{E}[a_n])\right] \\ \vdots & \ddots & \vdots \\ \mathbb{E}\left[(a_n-\mathbb{E}[a_n])(a_1-\mathbb{E}[a_1])\right] & \cdots & \mathbb{E}\left[(a_n-\mathbb{E}[a_n])(a_n-\mathbb{E}[a_n])\right]\end{bmatrix} \\
&= \begin{bmatrix}\operatorname{var}[a_1] & \operatorname{cov}[a_1,a_2] & \cdots & \operatorname{cov}[a_1,a_n] \\ \operatorname{cov}[a_2,a_1] & \operatorname{var}[a_2] & \cdots & \operatorname{cov}[a_2,a_n] \\ \vdots & \vdots & \ddots & \vdots \\ \operatorname{cov}[a_n,a_1] & \operatorname{cov}[a_n,a_2] & \cdots & \operatorname{var}[a_n]\end{bmatrix}
\end{aligned}
$$

(c) Show that the assumptions of homoskedasticity and no-autocorrelation of the error can be summarized as

$$
V[u|X] = \mathbb{E}[uu'|X] \quad = \quad \begin{bmatrix}\sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2\end{bmatrix} = \sigma^2 I_{n\times n} \tag{2}
$$

Solution:

$$
\begin{aligned}
V[u|X] = \mathbb{E}[(u-\mathbb{E}[u])(u-\mathbb{E}[u])'|X] = \mathbb{E}[uu'|X] &= \mathbb{E}\left[\begin{bmatrix}u_1\\ \vdots \\ u_n\end{bmatrix}\begin{bmatrix}u_1 & \dots & u_n\end{bmatrix}|X\right] \\
&= \begin{bmatrix}E[u_1^2|X] & E[u_1u_2|X] & \cdots & E[u_1u_n|X] \\ E[u_2u_1|X] & E[u_2^2|X] & \cdots & E[u_2u_n|X] \\ \vdots & \vdots & \ddots & \vdots \\ E[u_nu_1|X] & E[u_nu_2|X] & \cdots & E[u_n^2|X]\end{bmatrix}
\end{aligned}
$$

(d) Show that the matrix analog of (1) is

$$
\underset{k\times k}{V(\hat\beta|X)} = \begin{bmatrix}\operatorname{var}(\hat\beta_1|X) & & \\ & \ddots & \\ & & \operatorname{var}(\hat\beta_k|X)\end{bmatrix} = \sigma^2(X'X)^{-1}
$$

Solution: Recall that, assuming $y = X\beta + u$, we get

$$\hat{\beta} = (X'X)^{-1}X'y = \beta + (X'X)^{-1}X'u, \tag{3}$$

and that, assuming $\mathbb{E}(X'u) = 0$, it holds that

$$\mathbb{E}(\hat{\beta}|X) = \beta. \tag{4}$$

Hence, by (3) and (4)

$$\begin{aligned}
V[\hat{\beta}|X] =& \mathbb{E}\left[(\hat{\beta} - \mathbb{E}[\beta|X])(\hat{\beta} - \mathbb{E}[\beta|X])'|X\right] \\
=& \mathbb{E}\left[((X'X)^{-1}X'u)((X'X)^{-1}X'u)'|X\right] && \text{using (3)} \\
=& \mathbb{E}\left[((X'X)^{-1}X'u)(u'X(X'X)^{-1})|X\right] \\
=& \mathbb{E}\left[(X'X)^{-1}X'uu'X(X'X)^{-1}|X\right] \\
=& (X'X)^{-1}X'\mathbb{E}[uu'|X]X(X'X)^{-1} \\
=& (X'X)^{-1}X'\sigma^2 I_{n\times n}X(X'X)^{-1} && \text{given (2)} \\
=& \sigma^2(X'X)^{-1}
\end{aligned}$$

(e) Make an R script computing $\hat{V}[\hat{\beta}|X] = \hat{\sigma}^2(X'X)^{-1}$, where $\hat{\sigma}^2 = \frac{\sum_i \hat{u}_i^2}{n-k} = \frac{\hat{u}'\hat{u}}{n-k}$.

Solution: See uploaded ‘script PS2 OLS-Matrices’